Contents

Foreword

Preface

Model Syllabus for AI for Everyone

Acknowledgements

About the Authors

Chapter 1 Introduction

1.1 The Journey of Artificial Intelligence (AI)

1.1.1 The Early Years of AI: 1940s and 1950s

1.1.2 The Initial Journey: 1960s

1.1.3 First AI Winter: 1970s

1.1.4 Renewed AI Excitement: 1980s

1.1.5 AI in the 1990s

1.1.6 The Reinvention of AI: The 2000s Revolution

1.1.7 The AI Renaissance: Transforming the 2010s

1.1.8 And the AI Journey is on

1.2 What is AI?

1.2.1 Machine Learning: The Heart of AI

1.2.2 Pattern Recognition: The Power to Distinguish

1.2.3 Natural Language Processing (NLP): Conversations with Computers

1.2.4 Computer Vision: Seeing the World Through AI Eyes

1.3 Artificial General Intelligence (AGI)

1.3.1 Distinction from Narrow or Weak AI

1.4 Industry Applications of AI

1.4.1 Banking and Finance

1.4.2 Insurance

1.4.3 Healthcare

1.4.4 Retail and E-commerce

1.4.5 Manufacturing

1.4.6 Entertainment

1.4.7 Agriculture

1.4.8 Education

1.5 Challenges in AI

1.5.1 Data Privacy

1.5.2 Bias and Fairness

1.5.3 Transparency and Explainability

1.5.4 Ethical Concerns

1.5.5 Job Displacement

1.5.6 Regulation and Standards

1.5.7 Security Risks

1.5.8 Resource Intensiveness

1.5.9 Lack of Skilled Workforce

1.5.10 Interoperability

1.5.11 Accountability

1.6 Conclusion

1.7 Summary

1.8 Practice Exercises

1.8.1 Subjective Questions (10 Marks)

1.8.2 Subjective Questions (5 Marks)

1.8.3 Short Answer-type Questions (2 Marks)

1.8.4 Multiple-choice Questions (1 Mark)

1.9 Answer Keys

O'REILLY

Chapter 2 AI Subfields

2.1 Introduction

2.2 Computer Vision

2.2.1 Object Detection

2.2.2 Facial Recognition

2.2.3 Scene Understanding

2.2.4 Medical Imaging

2.2.5 Prominent Applications of Computer Vision

2.3 Natural Language Processing

2.3.1 Text Classification

2.3.2 Named Entity Recognition (NER)

2.3.3 Question Answering

2.3.4 Machine Translation

2.3.5 Text Generation

2.3.6 Text Summarization

2.4 Machine Learning

2.4.1 Classification

2.4.2 Regression

2.4.3 Clustering

2.4.4 Dimensionality Reduction

2.4.5 Recommendation Systems

2.4.6 Anomaly Detection

2.4.7 Deep Learning as a Special Case of Machine Learning

2.4.8 Reinforcement Learning

2.5 Robotics

2.5.1 Component of a Robot

2.5.2 Types of Robots

2.5.3 Sophia

2.6 Knowledge Engineering

2.6.1 Steps in Knowledge Engineering

2.6.2 Knowledge Representation

2.6.3 Fuzzy Sets and Logic

2.7 Summary

2.8 Practice Exercises

2.8.1 Subjective Questions (10 Marks)

2.8.2 Subjective Questions (5 Marks)

2.8.3 Short Answer-type Questions (2 Marks)

2.8.4 Multiple-choice Questions (1 Mark)

2.9 Answer Keys

2.9.1 Multiple-choice Questions

Chapter 3 Industrial Applications of AI

3.1 Application of AI in Healthcare

3.1.1 Role of AI in Medical Diagnosis

3.1.2 Role of AI in Early Detection and Disease Prevention

3.1.3 Role of AI in Drug Discovery and Development

3.1.4 AI-powered Virtual Medical Assistant (VMA)

3.1.5 AI-powered Robotics in Healthcare

3.1.6 Challenges of AI in Healthcare

3.2 Application of AI in Finance

3.2.1 AI in Algorithmic Trading

3.2.2 AI in Financial Risk Management

3.2.3 AI-based Customer Service

3.2.4 Challenges in Application of AI in Finance

3.3 Application of AI in Retail

3.3.1 Inventory and Store Layout Management

3.3.2 Personalized Shopping Experience

3.3.3 Customer Support

3.3.4 Challenges in Application of AI in Retail

3.4 Application of AI in Agriculture

3.4.1 Precision Farming

3.4.2 Crop Monitoring and Management

3.4.3 Smart Irrigation Systems

3.4.4 Challenges in the Application of AI in Agriculture

3.5 Application of AI in Education

3.5.1 Personalized Learning

3.5.2 Administrative Tasks

3.5.3 AI-based Language Processing Tools

3.5.4 Challenges in Application of AI in Education

3.6 Application of AI in Transportation

3.6.1 Traffic Management and Optimization

3.6.2 Ride-Sharing and Mobility Services

3.6.3 Safety and Security

3.6.4 Challenges in the Application of AI in Transportation

3.7 Conclusion

3.8 Summary

3.9 Practice Exercises

3.9.1 Subjective Questions (10 Marks)

3.9.2 Subjective Questions (5 Marks)

3.9.3 Short Answer-type Questions (2 Marks)

3.9.4 Multiple-choice Questions (1 Mark)

3.10 Answer Keys

3.10.1 Multiple-choice Questions

Chapter 4 Bias and Fairness in AI Systems

4.1 Introduction

4.2 Ethics in AI

4.3 Bias and Fairness in AI Systems

4.3.1 Types of Bias

4.3.2 Measuring and Detecting Bias

4.4 Transparency in AI Systems

4.4.1 Security Risks

4.4.2 Misinterpretation by Users

4.4.3 Competitive Disadvantage

4.5 Accountability of the AI Systems

4.6 Privacy and Data Protection Concerns

4.7 Security of the AI Models

4.8 Inclusivity in AI

4.9 Sustainability in AI

4.10 Robustness and Reliability

4.10.1 Concept Drift

4.10.2 Data Drift

4.11 Conclusion

4.12 Summary

4.13 Practice Exercises

4.13.1 Subjective Questions (10 Marks)

4.13.2 Subjective Questions (5 Marks)

4.13.3 Short Answer-type Questions (2 Marks)

4.13.4 Multiple Choice Questions

4.14 Answer Keys

4.14.1 Multiple-choice Questions

Chapter 5 AI in Research, Generative AI and Other Important Issues

5.1 AI In Experimentation and Multi-disciplinary Research

5.1.1 Application in Particle Physics (Identifying Particles in Collider Data)

5.1.2 Application in Astro Physics and Space

5.1.3 AI in Experimental Chemistry

5.1.4 AI in Biology

5.1.5 AI in Environmental Science

5.2 Generative AI Introduction

5.2.1 Chronology of the Different Developments

5.2.2 Synopsis of the Important Models

5.3 ChatGPT and Prompt Engineering

5.3.1 ChatGPT Context Setting

5.3.2 ChatGPT

5.3.3 Prompt Engineering

5.4 Emerging Trends and Future Directions in AI

5.5 AI and Social Inequality

5.6 Summary

5.7 Practice Exercises

5.7.1 Subjective Question (10 Marks)

5.7.2 Subjective Question (5 Marks)

5.7.3 Objective Question (2 Marks)

5.7.4 Multiple-choice Questions (1 Mark)

5.8 Answer Keys

5.8.1 Multiple-choice Questions

Model Question Paper

Index

Credits